

Lab: Autonomous Manipulation and Locomotion

Objective

Design autonomous action systems for manipulation and locomotion. You will implement grasp planning algorithms, whole-body control for mobile manipulation, and terrain-adaptive locomotion, framing each as Active Inference action selection.

Prerequisites

- Completion of the corresponding module.md lecture material
- Familiarity with Active Inference concepts (generative models, free energy, prediction errors)
- Basic pseudocode and diagram skills

Part 1: Requirements and Architecture

Define the requirements for an autonomous action system:

1. Specify the operational scenario (indoor warehouse, outdoor terrain, human-shared space).
2. Define performance requirements (speed, accuracy, reliability, safety).
3. Design the system architecture as a ROS2 node graph.
4. Map the architecture to Active Inference components (generative model, beliefs, actions, observations).

Write your response here

Part 2: Implementation Design

Design the core algorithms for the autonomous action system:

1. Specify the main algorithm in pseudocode.
2. Define the data structures and message types.
3. Specify timing constraints and computational budgets.
4. How does the system handle failures and edge cases?

Write your response here

Part 3: Integration and Testing

Design the integration and testing strategy:

1. Define unit tests for each component.
2. Design integration tests for the complete action pipeline.
3. Specify simulation-based testing scenarios.
4. Define metrics for evaluating autonomous action performance.

Write your response here

Part 4: Autonomy Evaluation

Evaluate the autonomy level of your system:

Autonomy Dimension	Your System	Ideal Autonomous System
Human intervention needed		
Failure recovery		
Novel situation handling		
Performance degradation		
Continuous operation time		

Write your response here

Summary Table

Concept	Classical Robotics	Active Inference	Your Design
Core mechanism			
Uncertainty handling			
Adaptation			
Key advantage			

References

- Lanillos, P., et al. (2021). Active Inference in Robotics and Artificial Agents. *Frontiers in Neurorobotics*.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press.
- Pio-Lopez, L., Nizard, A., Friston, K., & Pezzulo, G. (2016). Active Inference and robot control. *Journal of the Royal Society Interface*, 13(122).